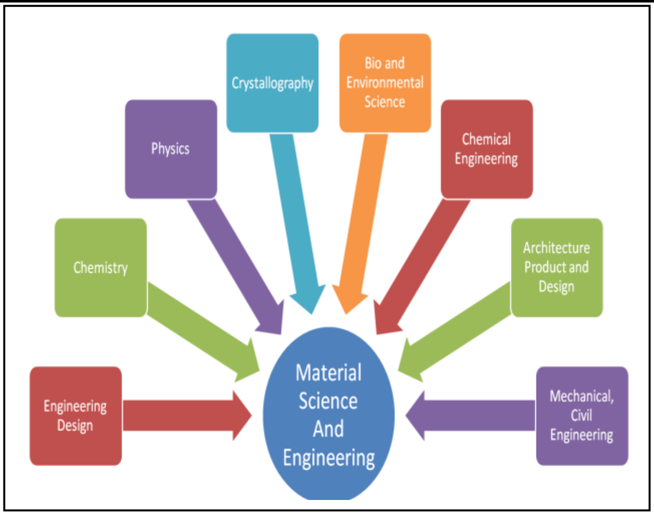
Module-IV

Material Science

Electrical Properties of Materials and Applications Electrical Conductivity in metals Resistivity and Mobility, Concept of Phonon, Matheissen's rule, Failures of Classical Free Electron Theory, Assumptions of Quantum Free Electron Theory, Fermi Energy, Density of States, Fermi Factor, Variation of Fermi Factor With Temperature and Energy. Numerical Problems.

Superconductivity Introduction to Super Conductors, Temperature dependence of resistivity, Meissner Effect, Critical Field, Temperature dependence of Critical field, Types of Super Conductors, BCS theory (Qualitative), Quantum Tunneling, High Temperature superconductivity, Josephson Junctions (Qualitative), DC and RF SQUIDs (Qualitative), Applications in Quantum Computing : Charge, Phase and Flux qubits, Numerical Problems



Review of Classical Free Electron theory (CFT): It was first proposed by 'Poul **Drude'**.Later,'H A **Lorentz'** improved this theory in certain aspects, hence, it is also known as '**Drude-Lorentz Theory**'.

According to this theory,the reason for conductivity in the conductors is due to presence of free electrons, which are moving freely and randomly in the entire volume of the conductorjust like gas atoms in a gas container. Based on this theory, the expression for electrical conductivity σ is obtained as,

$$\sigma = \frac{ne^2\tau}{m}$$

where, n is the number of free electrons(i.e., electron concentration), e is the charge of electron, τ is the relaxation time of free electrons and m is the mass of electron.

Failures of Classical FreeElectron theory:

Q: Explain the failures of classical free electron theory.**(4 marks)**

Though, Drude-Lorentz theory was successful in explaining the conductivity of metals but it had many failures.Among them following 2 failures may be considered.

1. *Temperature dependence of electrical conductivity:*

We know that the conductivity of a metal is proportional to inverse of temperature,

i.e., $\sigma \propto \frac{1}{T}$ (experimentally)

But, according to Drude-Lorentz theory, it is observed that:

$\sigma \propto \frac{1}{\sqrt{T}}$ (CFT)

This is not correct.Hence, temperature dependence could not be explained using CFT.

Poul Drude **(1863-1906)**
- German Physicist

Known for: Drude Model (Classical free electron theory),
- **Doctoral advisors:** Woldemar Voigt

H A Lorentz **(1853-1928)**
- Dutch Physicist

Known for:Classical free electron theory, Lorentz transformations, Lorentz force, theory of EM radiation, Lorentz matrix, Lorentz factor, etc
- **Awards:** Nobel Prize in Physics (1902)
- **Doctoral advisors:** Pieter Rijke

2. *Dependence of electrical conductivity on electron concentration (n):*

According to CFT, the conductivity is directly proportional to electron concentration 'n',

i.e., $$\sigma = \frac{ne^2\tau}{m} \Rightarrow \qquad \sigma \propto n$$

It indicates that the monovalent metals should have less conductivity compared to the bivalent or trivalent metals but a complete contradiction is observed experimentally, as follows:

Example	Copper (Monovalent)	Zinc (bivalent)	Aluminum (Trivalent)
	$n_{Cu} < n_{Zn} < n_{Al}$		
n	$8 \times 10^{28}\ m^{-3}$	$13 \times 10^{28} m^{-3}$	$18 \times 10^{28} m^{-3}$
According to **CFT** $\sigma_{Cu} < \sigma_{Zn} < \sigma_{Al}$	lesser	less	high
But, **experimentally** $\sigma_{Cu} > \sigma_{Zn} > \sigma_{Al}$	higher	high	less

Hence, the dependence of 'σ' on 'n' cannot be explained by CFT.

Quantum Free Electron Theory (QFT):

In order to resolve the failures of CFT, **Arnold Sommerfeld** developed the QFT by applying energy quantization and Pouli's exclusion principle.

Q: Give the assumptions of quantum free electron theory.**(4 marks)**

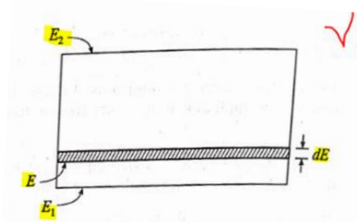
Assumptions:

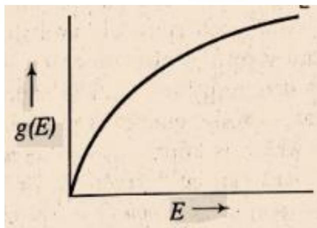
1. The energy values of the conduction electrons are ***quantized***.
2. The distribution of electrons in the various allowed energy levels occurs as per ***Pouli's exclusion principle.***
3. Free electrons travel in a constant potential inside the metal but stay confined within its boundaries.
4. The electrostatic electron-ion attractions and electron-electron repulsions are negligible.
5. The electrons are considered as wave like particles.

Density of States:

Def: "It is the number of available states per unit energy range centered at a given energy ***E*** in the valence band of a material of unit volume".

It is a mathematically continuous function and product **g(E)·dE** gives the number of states in the energy interval **dE**,where **g(E)** is density of states function.





Consider a material of unit volume. Let energy band be spread in the energy interval between E_1 and E_2, and dE be the infinitesimally small interval in **E,**then the number of energy states in the range E and (E+dE) is obtained by evaluating the ***g(E)dE.***

By Quantum mechanical calculation, it can be shown that,

$$g(E)dE = \left(\frac{8\sqrt{2}\pi m^{3/2}}{h^3}\right) E^{1/2} dE$$

$\Rightarrow$ Density of states is proportional to square root of energy value, i.e., $g(E)dE \propto \sqrt{E}$

Q: Give a brief account for Fermi-Dirac distribution theory.**(4 marks)**

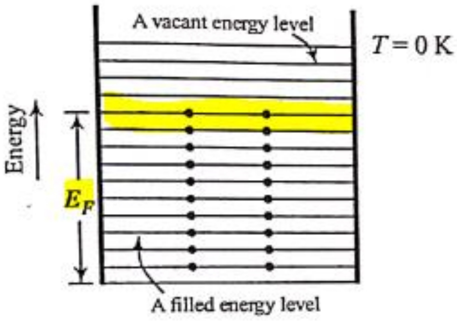
Fermi-Dirac statistics:

"Fermi –Dirac statistics is the statistical rule applied to the distribution ofidentical, indistinguishable particles of spin 1/2 called **fermions** like **electrons,**which obey **Pauli's exclusion principle**".

Thus, F-D statistics explains about how electrons orderly distributed in various energy states under thermal equilibrium.

Fermi energy (E_F) and Fermi level:

We know that as per Pauli's exclusion principle, each energy level can accommodate a maximum of two electrons- one with spin up and other one with spin down. The filling up of these electrons in various energy levels is undertaken from lowest level to higher levels as shown in **Fig**.



Now we define Fermi energy and level as follows.

"The highest energy occupied by an electron at zero degree Kelvin is called Fermi energy (E_F)and the corresponding energy level is referred as Fermi level".

F-D statistics permits the evaluation of probability of finding electronsoccupying energy levels through a function called **Fermi Factor [*f(E)*].**

Fermi Factor [*f(E)*]:

Q: Define Fermi factor. Explain the variation of Fermi factor with temperature.**(6 M)**

Fermi factor evaluates probability of finding electrons under thermal excitations. Though, thermal excitationsare random, but the resulting distribution of electrons in various energy levels after excitation is not random but a systematic one. Fermi factor is defined as,

"Fermi factor is the probability of occupation of given energy state by an electron in the material at the thermal equilibrium".

The probability (***f(E)***) of finding electron occupying an energy level ***E*** at a temperature ***T***, is given by,

$$f(E) = \frac{1}{e^{(E-E_F)/kT} + 1}$$

Enrico Fermi(1901-1954)

- Italian Physicist, later American Physicist

Known for: F-D statistics, Fermi Golden Rule, Nuclear chain reaction, β-decay, etc.

- **Awards:** Nobel prize in 1938
- **Doctoral advisors:** Max Born, Paul Ehrenfest,

Paul Dirac (1902-1984)

- British theoretical Physicist

Known for:F-D statistics, Dirac algebra, delta function etc

- **Awards:** Nobel Prize in Physics (1933)
- **Doctoral Students:** Homi J. Bhabha, etc.

Dependence of Fermi Factor on Temperature and effect on occupancy of Energy Levels:

Case (i): *Probability of occupation at T=0 K for E<E_F*

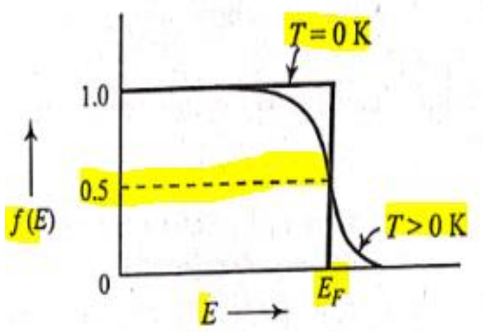
Under this case, we have for the probability,

$$f(E) = \frac{1}{e^{(E-E_F)/kT}+1} \Rightarrow \qquad f(E) = \frac{1}{e^{-\infty}+1} = \frac{1}{0+1} \qquad |\because \text{(E-E}_F) \to \text{-ve\& T = 0 K}$$

$\therefore$ $$f(E) = 1$$

Here, $f(E) = 1$ means the energy level is certainly occupied and ***E<E_F*** applies to all the energy levels below ***E_F*** (see **Fig).**

∴ At ***T=0 K,*** all the energy levels below the **Fermi level** are occupied.



Case (ii): *Probability of occupation at T=0 Kfor E>E_F*

Here, we have for the probability,

$$f(E) = \frac{1}{e^{(E-E_F)/kT}+1} \Rightarrow \qquad f(E) = \frac{1}{e^{\infty}+1} = \frac{1}{\infty+1} \qquad |\because \text{(E-E}_F) \to \text{+ve\& T = 0 K}$$

$\therefore$ $$f(E) = 0$$

Here, $f(E) = 0$ means the energy level is unoccupied and $E > E_F$ applies to all the energy levels above E_F

∴ At ***T= 0 K,*** all the energy levels above the **Fermi level** are unoccupied.

Hence, at ***T=0 K*** the variation of ***f(E)*** for different energy values becomes a step function as shown in **Fig**.

Case (iii): ***Probability of occupation at T >0 K(i.e., at ordinary temperatures)***

At ordinary temperatures, though the value of $f(E)$ remains **1** for $E << E_F$,it starts decreasing from **1** as the value of ***E*** becomes closer to E_F(see the curve for T > 0 K in **Fig)**.

The values of $f(E)$becomes ½ at $E = E_F$. This is because,

$$f(E) = \frac{1}{e^{\left(\frac{E-E_F}{kT}\right)}+1} \Rightarrow \qquad f(E) = \frac{1}{e^0+1} = \frac{1}{1+1} \quad |\because (E-E_F) \to 0 \ \& \ T \neq 0 \text{ K}$$

$$\therefore \qquad f(E) = \frac{1}{2} \text{ or } \ 0.5$$

Further, for $E > E_F$, the probability $f(E)$value falls off to zero rapidly.

Fermi Temperature (T_F):*"It is the temperature at which the average thermal energy of the free electrons in metal becomes equal to the Fermi energy at 0 K".*

i.e., $$kT_F = E_F \Rightarrow T_F = \frac{E_F}{k}$$

where, ***k*** is Boltzmann's constant, E_F is Fermi energy.

Note: Fermi Temperature is only a theoretical concept, since at ordinary temperature, it is not possible for the electrons to receive the thermal energy in a magnitude of E_F.

For Eg: Let us consider the case in which, $E_F = 3\ eV,$ then $T_F = \frac{E_F}{k} = \frac{3\times1.602\times10^{-19}}{1.381\times10^{-23}} \Rightarrow T_F = 34800\ K$

This is quite exaggerated temperature to be realized in practice.

Fermi Velocity (V_F): *"The velocity with which electrons occupy the Fermi level is called Fermi velocity"*.

i.e., $$E_F = \frac{1}{2}mv_F^2 \Rightarrow V_F = \sqrt{\frac{2E_F}{m}}$$

Expression for Electrical Conductivity (σ) based on Quantum Free Electron theory (no derivation)**:**

Sommerfeld arrived at an expression for electrical conductivity of metals based on QFT, as:

$$\sigma = \frac{ne^2}{m^*}\left(\frac{\lambda}{V_F}\right)$$

where, m^* is the effective mass of the electron, λ is mean free path of electrons, V_F is Fermi velocity, ***n*** is concentration of electrons.

Effective mass (m^*): When a metal is subjected to the influence of an electric field then the mass possessed by the electron which moves under the combined influence of applied electric field and that of a periodic potential due to lattice ions is called effective mass. It is different from its true mass.

Success of Quantum Free Electron Theory:

Q: Discuss two successes of quantum free electron theory.**(04 marks)**

QFT has been successful in explaining the following facts:

1. **Temperature dependence of electrical conductivity**:

According to quantum free electron theory, we know that, $\sigma = \frac{ne^2}{m^*}\left(\frac{\lambda}{V_F}\right) \Rightarrow \sigma \propto \lambda$ (1)

As per QFT, the temperature dependence of $\boldsymbol{\sigma}$will come from the nature of dependence of $\boldsymbol{\lambda}$on $\boldsymbol{T}$. Due to $\boldsymbol{T}$**,** lattice ions vibrate in all the directions equally, creating a circular cross-section of area $\boldsymbol{\pi r^2}$($\boldsymbol{r}$; amplitude of vibration). $\boldsymbol{\lambda}$ and this area related as**,** $\boldsymbol{\lambda} \propto \frac{\mathbf{1}}{\boldsymbol{\pi r^2}}$. But, $r^2 \propto T \Rightarrow \boldsymbol{\lambda} \propto \frac{\mathbf{1}}{\boldsymbol{T}}$ (2)

Comparing **eqn (1**) and **(2**), we have

$$\boldsymbol{\sigma} \propto \frac{\mathbf{1}}{\boldsymbol{T}}$$

This is in good agreement with experimental results.

2. Dependence of electrical conductivity on electron concentration:

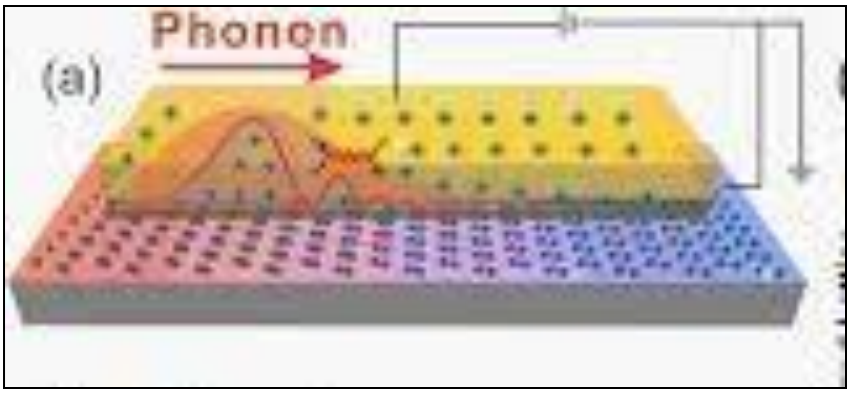
According to QFT, we know that, $\sigma = \frac{ne^2}{m^*}\left(\frac{\lambda}{V_F}\right)$

From this equation it is clear that the value of$\boldsymbol{\sigma}$ depends not only on $\boldsymbol{n}$ but also on ratio $\left(\frac{\boldsymbol{\lambda}}{\boldsymbol{V_F}}\right)$ and $\boldsymbol{m^*}$**.**

If we compare the cases of copper (monovalent) and aluminum (trivalent):-Though, the value of $\boldsymbol{n}$ for Al is **2.13** times greater than that of **Cu,**but the value of $\left(\frac{\boldsymbol{\lambda}}{\boldsymbol{V_F}}\right)$ for **Cu** is about **3.73** times higher than that of **Al.** Hence conductivity of **Cu** exceeds that of **Al**(i.e., $\boldsymbol{\sigma_{Cu}} > \boldsymbol{\sigma_{Al}}$).

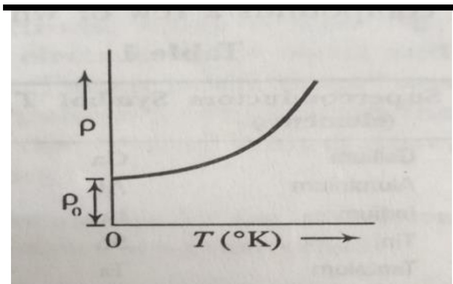
Also value of $\boldsymbol{m^*}$ for Al is **1.08** times that of Cu.Because of inverse dependence of $\boldsymbol{\sigma}$ on $\boldsymbol{m^*}$**,** $\boldsymbol{\sigma}$ of Cu becomes higher than the $\boldsymbol{\sigma}$of Al (i.e., $\boldsymbol{\sigma_{Cu}} > \boldsymbol{\sigma_{Al}}$).

Concept of phonon:



All the metals have crystalline structure. Hence atoms or ions are situated at particular position of crystal lattice. Because of thermal energy these ions are vibrating in all directions from their mean position. These vibrations of ions in lattice are quantised. The quantised lattice vibrations are called phonons. These are the particle like entities which carries the unit energy of elastic field in a particular mode. The energy of the phonon is given by hʋ. The free electrons collide with ion core at lattice point. The collision results in a change in the direction of velocity of the electrons. In absence of electric field, the velocities of electrons are in random directions due to which there is no net transfer of charge. According to Drude –Lorentz theory all metals contain free electrons move through the positive ionic core of the metals. The metal is then pictured to be held together by electrostatic forces of attraction between the positively charged ions and negatively charged electron gas. Mutual repulsion between negative electrons is ignored in this theory.

Matthiessen’s rule:



DEPENDENCE OF ρ ON T
Fig. 1

The resistivity of metal is due to scattering of conduction electrons by lattice vibrations. Hence electrons move under the action of electric field. The amplitude of lattice vibration is proportional to temperature. If the temperature of the material increases, amplitude of lattice vibration also increases, then more electrons are scattered which leads to increase in the resistivity of the material.

The resistivity ρ of a metal containing small amount of impurities may be written as:

$$\boldsymbol{\rho = \rho_0 + \rho(T)} \qquad (1)$$

Where ρ_0 is a constant which increases with increasing impurity content and ρ (T) is the temperature –dependent part of the resistivity.

Equation (1) is known as the **Matthiessen's rule**

In a conductor, the resistivity decreases with temperature and reaches minimum value at T=0. This non-zero resistivity is due to presence of impurities in the metal and it is called residual resistivity $\boldsymbol{\rho_0}$.

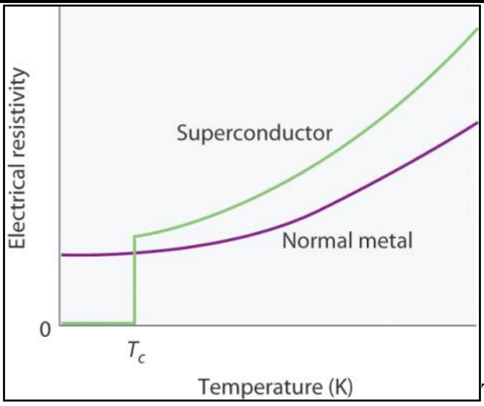
SUPERCONDUCTIVITY

Superconductivity is a phenomenon observed in some metals. There is disappearance of electrical resistivity at temperature approaching 0K. The phenomenon of superconductivity was discovered by Kamerlingh Onne. He was studying properties of Hg at very low temperature, he found that resistivity of Hg suddenly dropped to 0 at 4.2K. That is Hg converted into superconducting material at 4.2K. At very low temperature, a normal conductor has some resistivity but a superconductor resistivity suddenly drops to zero. Some good electrical conductors like copper and gold are not good superconductors. On other hand materials like Zinc and lead are not good electrical conductors, but they are good superconductors. The superconducting materials show their superconducting behavior below certain temperature known as critical temperature T_c. They can carry very high currents without loss of energy and their conductivity is infinite. The ability to achieve very high currents in electromagnet makes it possible to produce very high magnetic field. When temperature of superconducting material is increased material transform into normal conductor above critical temperature T_c

Definition of superconductivity and Critical Temperature:

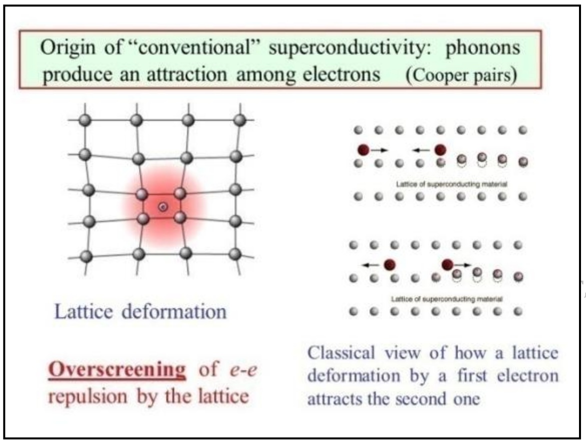
The resistance offered by certain materials to the flow of electric current abruptly drops to zero below a threshold temperature. This phenomenon is called superconductivity and the threshold temperature is called critical temperature.

Temperature Dependence of Resistivity of Superconductor :



The dependence of resistivity ρ of a superconductor on temperature is shown in Fig. resistivity in the non-superconducting state (normal metal) decreases with decrease in temperature as in the case of a normal metal up to a particular temperature T_c. At T_c ρ abruptly drops to zero. T_c is called the critical temperature and signifies the transition from normal state to the superconducting state of the material under study. The critical temperature is different for different superconductors. For ex. Mercury loses its resistance completely and turns into a superconductor at 4.2K.

BCS (Barden Cooper and Schrieffer's Theory) Theory

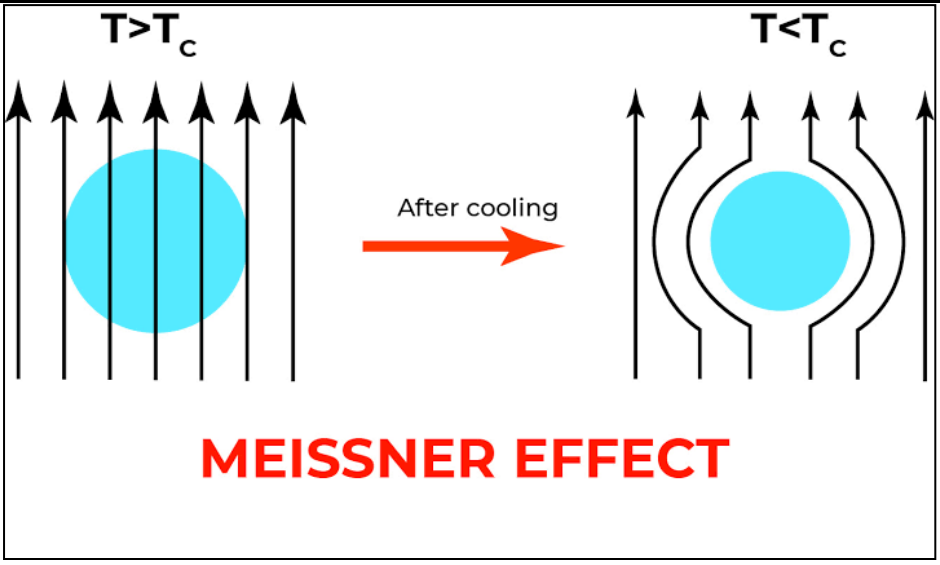


- The indirect interaction proceeds when one electron interacts with lattice and deforms it and second electron reaches deformed lattice adjust itself to take advantage of deformation to lower its energy. Thus the second electron interacts with first electron via lattice deformation forms cooper pair. The cooper pair is a bound pair of electrons formed by interaction between electrons with opposite spin and momenta in phonon field.
- An attractive interaction between electrons leads to separation of ground state from excited state by an energy gap. The critical field, thermal properties and most of the electromagnetic properties are consequences of energy gap.
- The penetration depth and coherence length emerge as a natural consequence of B.C.S Theory. The Landon equation is obtained for magnetic field that vary slowly in space. Thus meissner effect can be obtained in natural way.
- The criterion for critical temperature in element involves electron density at Fermi level and electron lattice interaction which can be estimated from electrical resistivity measurement. Because resistivity at room temperature is a measure of electron-phonon interaction.
- The magnetic flux through a superconducting ring is quantized and the effective unit of charge is 2e rather than e. the BCS ground state involves pairs of electrons thus flux quantization in terms of pair charge 2e is a consequence of theory

Meissner Effect (Effect of Magnetic Field):

A superconducting material kept in a magnetic field expels the magnetic flux out of its body when it is cooled below the critical temperature T_c and thus it becomes a perfect diamagnet. This phenomenon is called 'Meissner effect'.

Demonstration of Meissner Effect:



Meissner Effect is the expulsion of the magnetic field, which converts a material from a normal state to a superconducting state when cooled down below the critical temperature.

Meissner Effect is the effect that converts a material from a normal state to a superconducting state. When a material is called below critical temperature at which the magnetic field from the interior of the material is exploded, then this material is transformed into the superconductor.

The temperature required to convert the material into the superconductor is called the critical temperature. This transition temperature is near absolute temperature (0K), This superconductor material has zero electric resistance. When a magnetic field is applied at the higher value of temperature $T>T_c$, the superconductor has a minor effect. These magnetic fields passed through the superconductor quickly. But when $T<T_c$, the magnetic field is thrown out from inside the superconductor and bends on all sides of it when the magnetic field is applied.

This expulsion of the magnetic field is the generation of magnetisation inside the superconductor due to no resistance flow of surface currents. This generated diamagnetic magnetization is inverse and opposite to the applied magnetic field, which neutralises the magnetic field around the superconductor.

The Meissner Effect suggests that the value of the magnetic field inside a superconductor is zero.

$B=0$

The magnetic induction inside the specimen for the normal conditions is given by the following equation.

$B=\mu_0(H+I)$

Here, I is the magnetization which is generated inside the specimen and H is the magnetic field applied externally.

By both equations, we have,

$\mu_0 (H+I) = 0$

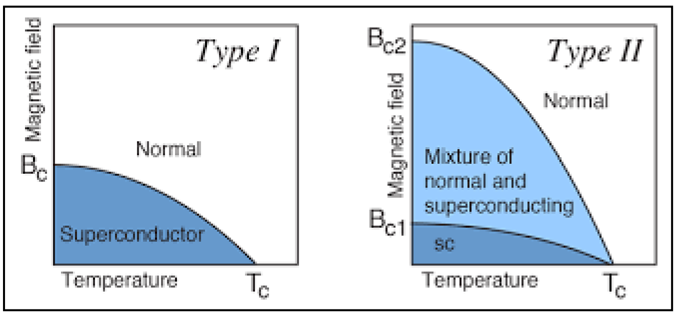
$(H+I) = 0$

$H=-I$

$$\chi = \frac{I}{H} = -1$$

The diamagnetic material susceptibility is equal to -1, so the material is ideally diamagnetic.

Types of super conductors:



Type I Superconductors: These superconductors exhibit complete Meissner effect. In the presence of an external magnetic field $H<H_c$, the material in superconducting state is a perfect diamagnet. Since it is a diamagnet it possess negative magnetic moment(-M).

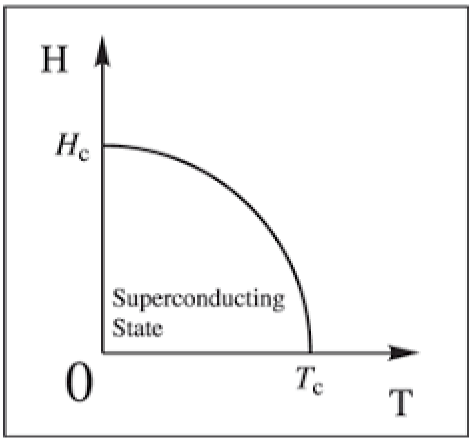
The dependence of magnetic moment on H for Type I superconductors is shown in Fig. As soon as the applied field H exceeds H_c, the entire material becomes normal by loosing its diamagnetic property completely, and the magnetic flux penetrates throughout the body. The resistance changes from zero to a value as applicable to a normal conductor. The critical field value for Type I superconductors are found to be very low. The low H_c value excludes them from being used for construction of superconducting magnets wherein they are exposed to extremely high magnetic fields.

1. **Type II Superconductors**

A Type II superconductor is characterized by two critical magnetic fields H_{c1} , and H_{c2}. For any applied field strength less than the critical value H_{c1}, it expels the magnetic field from its body completely, and behaves as a perfect diamagnet from end to end. When H exceeds H_{c1}, the flux penetrates the body and fills in partially. With further increase in H, the flux filling also increases and covers the entire body when H becomes equal to or greater than a second critical value H_{c2}. The material then turns into a normal conductor. H_{c1} andH_{c2} are called the lower critical field and the upper critical field respectively.

The dependence of magnetic moment on the external magnetic field is shown in Fig.. When the applied field strength is between H_{c1} and H_{c2}, the material is in a mixed state. This state is called Vortex (mixed) state. The material doesn't show complete meissner effect. The flux penetration occurs through small channelized parts of the body called filaments. They are the bits of material which had turned to normal state from the superconducting state when $H>H_{c1}$. These filaments increase in number with increase in H till $H=H_{c2}$ when they spread into the entire body, and the material as a whole returns to its normal conducting state.

Temperature dependence of critical field:



The minimum strength of applied magnetic field required to change the superconducting state to normal state below critical temperature is called **Critical field (H_c).**

For a superconducting material, the critical field will be higher when temperature is lower. If T is temperature of superconducting material ($T<T_c$), H_c the critical field and H_0 the field required to turn the superconductor to normal conductor at 0^0K. Then the relation for critical field is given by

$H_c = H_0\left[1-\frac{T^2}{T_c^2}\right]$ The dependence of H_c on T is given in the fig.

Under the influence of magnetic field whose strength is greater than H_c the material can never become superconducting however low the temperature may be.

High-temperature superconductors:

All high temperature superconductors are different types of oxides of copper and exhibits particular type of crystal structure called perovskite crystal structure. It is interesting to note that critical temperature is higher for which have more number copper oxygen layers in structural units.

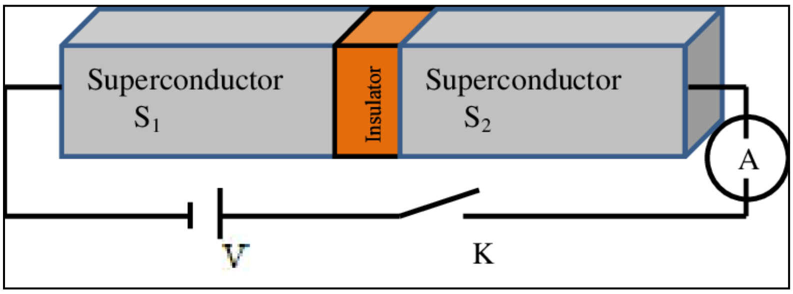
The formation of super currents in high temperature superconductors is direction dependent super-currents are strong in the copper-oxygen planes and they are weak in the direction perpendicular to planes.

High temperature superconductors are defined as materials that behave as superconductors at temperatures above 77 K. . In absolute terms, these "high temperatures" are still far below ambient, and therefore require cooling. The first high-temperature superconductor was discovered in 1986, by IBM researchers Bednorz and Müller. Most of the high-temperature superconductors are type-II superconductors.

The major advantage of high-temperature superconductors is that they can be cooled by using liquid nitrogen, as opposed to the previously known superconductors which require expensive and hard-to-handle coolants. A second advantage of high-T_c materials is they retain their superconductivity in higher magnetic field, This is important when constructing superconducting magnets, a primary application of high-T_c materials.

The majority of high-temperature superconductors are ceramic materials. Ceramic superconductors are suitable for some practical uses but they still have many manufacturing issues. For example, most ceramics are brittle which makes the fabrication of wires from them very problematic.

Josephson Effect:



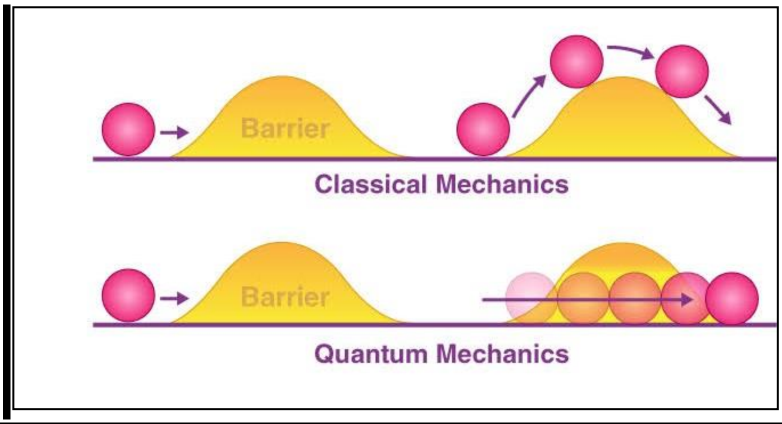
The current flows across the two superconductors separated by an insulating layer in the absence of electric and magnetic field is called **Josephson effect.** . It is an example of a macroscopic quantum phenomenon, The Josephson effect has many practical applications because it exhibits a precise relationship between different physics quantities, such as voltage and frequency, facilitating highly accurate measurements.

The Josephson effect produces a current, known as a super-current, that flows continuously without any voltage applied, across a device known as a Josephson junction (JJ). These consist of two or more superconductors coupled by a weak link.

Josephson junction:

A Josephson junction is a quantum mechanical device which is made of two superconducting electrodes separated by a barrier (thin insulating tunnel barrier, normal metal, semiconductor, etc.). A π Josephson junction is a Josephson junction in which the Josephson phase φ equals π in the ground state, i.e. when no external current or magnetic field is applied. The super-current I_s through a Josephson junction (JJ) is generally given by $I_s = I_c \sin(\varphi)$, where φ is the phase difference of the superconducting wave functions of the two electrodes and I_c is the critical current i.e maximum super-current that can exist through the Josephson junction.

Quantum Tunnelling:



A quantum tunnelling or barrier penetration, or simply tunnelling is a quantum mechanical phenomenon in which an object such as an electron or atom passes through a potential energy barrier that, according to classical mechanics, the object does not have sufficient energy to enter.

Tunneling is a consequence of the wave nature of matter, where the quantum wave function describes the state of a particle or other physical system, and wave equations such as the Schrödinger equation describe their behaviour. The probability of transmission of a wave packet through a barrier decreases exponentially with the barrier height, the barrier width, and the tunneling particle's mass. So tunneling is seen most prominently in low-mass particles such as electrons or protons. Tunneling is readily detectable with barriers of thickness about 1–3 nm or smaller for electrons, and about 0.1 nm or smaller for heavier particles such as protons or hydrogen atoms. Some sources describe the mere penetration of a wave function into the barrier, without transmission on the other side, as a tunneling effect.

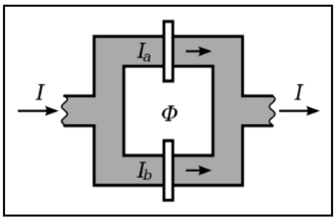
SQUID:

A **SQUID** (superconducting quantum interference device) is basically a sensitive magnetometer. The heart of squid is a superconducting ring which contains one or more Josephson junction. It is used to measure extremely subtle magnetic fields, based on superconducting loops containing Josephson junctions.

SQUIDs are sensitive enough to measure fields as low as 5×10^{-14} T with a few days of averaged measurements. Their noise levels are very low

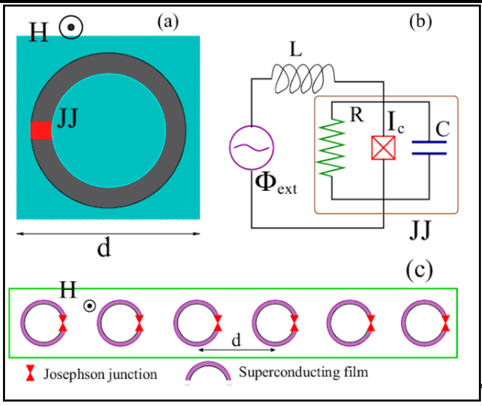
There are two main types of SQUIDs: a direct current (DC) SQUID and radio frequency (RF) SQUID. RF SQUIDs can work with only one Josephson junction (superconducting tunnel junction), which might make them cheaper to produce, but are less sensitive.

DC SQUID:



When the Josephson junctions are sandwiched between two insulating layers in the form of ring, it forms DC squid. The DC SQUID has two Josephson junctions in parallel in a superconducting loop. It is based on the DC Josephson effect. In the absence of any external magnetic field, the input current splits into the two branches each with currents I_a and I_b equally hence it gives the phase difference zero. . If a small external magnetic field is applied to the superconducting loop, phases of currents changes, then wavefunctions at the junction interfere. then a screening current, I_s begins to circulate the loop that generates the magnetic field cancelling the applied external flux, and creates an additional Josephson phase which is proportional to this external magnetic flux. The induced current is in the same direction as I in one of the branches of the superconducting loop, and is opposite to I in the other branch; the total current becomes $\frac{I}{2}+I_s$ in one branch and $\frac{I}{2}-I_s$ i in the other. As soon as the current in either branch exceeds the critical current I_c of the Josephson junction, a voltage appears across the junction. Since the current-voltage characteristic is hysteretic, a resistance R is connected to remove hysteresis.

RF SQUID:



The RF SQUID is composed of one Josephson junction and RF coil. RF coil is used to readout changes due to change in flux. It is based on the AC Josephson effect and uses only one Josephson junction. Suppose one junction superconducting loop is put in an external flux, in one junction superconducting loop is coupled to a circuit driven by rf current source.

The tuned circuit is driven by constant rf oscillator which is weakly coupled to the loop. Measuring the change in input coil current is done by counting number of periods the coil produces in the detected rf output. Because the detected rf output is periodic function. It is less sensitive compared to DC SQUID but is cheaper and easier to manufacture in smaller quantities. The RF SQUID is inductively coupled to a resonant tank circuit. Depending on the external magnetic field, as the SQUID operates in the resistive mode, the effective inductance of the tank circuit changes, thus changing the resonant frequency of the tank circuit. These frequency measurements can be easily taken, and thus the losses which appear as the voltage across the load resistor in the circuit are a periodic function of the applied magnetic flux with a period of ϕ_0.

Applications of Superconductivity in Quantum Computing:

Charge Qubit:

In quantum computing, a **charge qubit** (also known as **Cooper-pair box**) is a qubit whose basis states are charge states (i.e. states which represent the presence or absence of excess Cooper pairs in the island). In superconducting quantum computing, a charge qubit is formed by a tiny superconducting island coupled by a Josephson junction to a superconducting reservoir. The state of the qubit is determined by the number of Cooper pairs which have tunneled across the junction. In contrast with the charge state of an atomic or molecular ion, the charge states of such an "island" involve a macroscopic number of conduction electrons of the island. The quantum superposition of charge states can be achieved by tuning the gate voltage *U* that controls the chemical potential of the island. The charge qubit is typically read-out by electrostatically coupling the island to an extremely sensitive electrometer such as the radio-frequency single-electron transistor.

Phase Qubit:

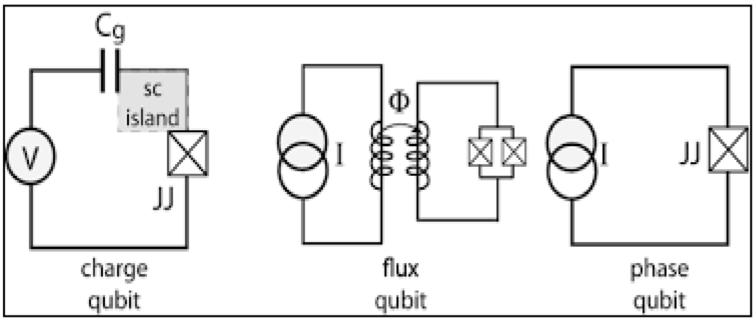
In superconducting quantum computing, the **phase qubit** is a superconducting device based on the superconductor–insulator–superconductor (SIS) Josephson junction, designed to operate as a quantum bit, or qubit.

The phase qubit is closely related, yet distinct from, the flux qubit and the charge qubit, which are also quantum bits implemented by superconducting devices. The major distinction among the three is the ratio of Josephson energy vs charging energy (the necessary energy for one Cooper pair to *charge* the total capacitance in the circuit):

- For phase qubit, this ratio is on the order of 10^6, which allows for macroscopic bias current through the junction;
- For flux qubit it's on the order of 10, which allows for mesoscopic supercurrents
- For charge qubit it's less than 1, and therefore only a few Cooper pairs can tunnel through and charge the Cooper-pair box.

Flux Qubit:

In superconducting quantum computing, **flux qubits** (also known as **persistent current qubits**) are micrometer sized loops of superconducting metal that is interrupted by a number of Josephson junctions. These devices function as quantum bits. During fabrication, the Josephson junction parameters are engineered so that a persistent current will flow continuously when an external magnetic flux is applied. Only an integer number of flux quanta are allowed to penetrate the superconducting ring, resulting in clockwise or counter-clockwise supercurrents in the loop to compensate a non-integer external flux bias. When the applied flux through the loop area is close to a half integer number of flux quanta, the two lowest energy eigenstates of the loop will be a quantum superposition of the clockwise and counter-clockwise currents. The two lowest energy eigenstates differ only by the relative quantum phase between the composing current-direction states.



Solved Examples (Conductors)

Example 1.

FERMI ENERGY, FERMI FACTOR AND FERMI-DIRAC DISTRIBUTION

Example 2 :

The Fermi level in silver is 5.5 eV. Find the veloci conduction electrons in silver. V.T.U. D

Calculate the Fermi velocity and the mean free for the conduction electrons in silver, given that its Fermi er is 5.5 eV, and the relaxation time for electrons is 3.83 × 10 V.T.U. J

Data :

Fermi energy, $E_F = 5.5 \text{ eV} = 5.5 \times 1.602 \times 10^{-19}$ J.

Relaxation time for electrons, $\tau = 3.97 \times 10^{-14}$ s.

To Find :

Fermi velocity, $v_F = ?$

Mean free path, $\lambda = ?$

Solution :

We have,

$$v_F = \sqrt{\frac{2E_F}{m}},$$

$$= \sqrt{\frac{2 \times 5.5 \times 1.602 \times 10^{-19}}{9.11 \times 10^{-31}}},$$

$$= 1.39 \times 10^6 \text{ m/s}.$$

Mean free path is given by,

$$\lambda = v_F \tau,$$

$$\therefore \quad \lambda = 1.39 \times 10^6 \times 3.83 \times 10^{-14} = 5.518 \times 10^{-8}$$

Example 2.

Example 6 :

Calculate the probabilities of an electron occupying an energy level 0.02 eV above the Fermi level and that in an energy level 0.02 eV below the Fermi level at 200 K.

V.T.U(O.S), June 15, V.T.U., Jan. 11

Solution :

For the case of electron occupying the energy level 0.02 eV above E_F, at 200 K,

please refer to the previous example. The answer is,

$$f(E) = 0.24 \text{ at } 200 \text{ K}. \quad \text{......(1)}$$

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Now, for the case of electron, occupying the energy level 0.02 eV below E_F, at 200 K,

we have, $\Delta E = E_F - E = 0.02$ eV.

or, $E - E_F = -0.02 \text{ eV} = 0.02 \times 1.6 \times 10^{-19}$ J .

$$\therefore \quad f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1} = \frac{1}{e^{\frac{0.02 \times 1.6 \times 10^{-19}}{1.38 \times 10^{-23} \times 200}} + 1}$$

$$\text{or,} \quad f(E) = \frac{1}{e^{-1.1594} + 1} = \frac{1}{0.3137 + 1}$$

or, $f(E) = 0.76.$

∴ $f(E) = 0.24$ for $E = 0.02$ above E_F,

and, $f(E) = 0.76$ for $E = 0.02$ below E_F.

Example 3.

Example 3 :

The Fermi level in silver is 5.5 eV at Zero Kelvin. Calculate the number of free electrons per unit volume and the probability of occupation for electrons with energy 5.6 eV in silver at the same temperature. (Given: 1 eV = 1.602×10^{-19} J). Manipal Univ. July 01

Data :

For silver, $E_{F_0} = 5.5 \text{ eV} = 5.5 \times 1.602 \times 10^{-19}$ J,

$= 8.811 \times 10^{-19}$ J.

To Find :

The free electron concentration, $n = ?$

At $T = 0$, and for $E = 5.6$ eV, $f(E) = ?$

Solution :

We have the equation,

$$E_{F_0} = \left(\frac{h^2}{8m}\right)\left(\frac{3}{\pi}\right)^{2/3} n^{2/3}.$$

After substituting the values, we have,

$$8.811 \times 10^{-19} = \frac{\left(6.63 \times 10^{-34}\right)^2}{\left(8 \times 9.11 \times 10^{-31}\right)}\left(\frac{3}{\pi}\right)^{2/3} n^{2/3},$$

$$= 5.85 \times 10^{-38} \times n^{2/3},$$

$$\therefore \quad n^{2/3} = \frac{8.811 \times 10^{-19}}{5.85 \times 10^{-38}} = 1.506 \times 10^{-19}.$$

$$\therefore \quad n = \left(1.506 \times 10^{19}\right)^{3/2},$$

$$= 5.84 \times 10^{28} / m^3.$$

∴ The number of free electrons/unit volume in silver is $5.85 \times 10^{28}/m^3$.

Now $E = 5.6$ eV and $E_F = 5.5$ eV (by data).

$$\therefore \quad E > E_F.$$

But at $T = 0$, no energy levels are occupied above E_F.

$$\therefore \quad f(E) = 0.$$

Example 4.

Example 5 :

Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K and 400 K in a material. V.T.U. Jan 10, July 06, July 02

Data :

$$E - E_F = 0.02 \text{ eV} = 0.02 \times 1.6 \times 10^{-19} \text{ J}.$$

To find :

(i) $f(E)$ at 200 K = ?
(ii) $f(E)$ at 400 K = ?

Solution :

(i) *Evaluation of $f(E)$ at 200 K :*

We have,

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1},$$

$$= \frac{1}{e^{\frac{0.02 \times 1.6 \times 10^{-19}}{1.38 \times 10^{-23} \times 200}} + 1} = \frac{1}{e^{1.1594} + 1},$$

$$= \frac{1}{3.188 + 1} = 0.24.$$

$$\therefore \quad f(E) = 0.24 \text{ at } 200 \text{ K}.$$

(ii) *Evaluation of $f(E)$ at 400 K :*

$$f(E) = \frac{1}{e^{\frac{0.02 \times 1.6 \times 10^{-19}}{1.38 \times 10^{-23} \times 400}} + 1}$$

$$= \frac{1}{e^{0.5797} + 1} = \frac{1}{1.7855 + 1} = 0.36,$$

$$\therefore \quad f(E) = 0.36 \text{ at } 400 \text{ K}.$$

Example 5.

Example 4 :

Calculate the Fermi energy in eV for a metal at Zero Kelvin, whose density is 10500 kg/m^3, atomic weight is 107.9, and it has one conduction electron per atom. (Given : 1 J = 6.24×10^{18} eV and $N_A = 6.025 \times 10^{26}$/k mole) V.T.U., Jan 06, July 03, Kuvempu Univ. March 2000

Data :

Density of the given metal, $D = 10500$ kg/m^3.

Atomic weight of the given metal, $A = 107.9$.

Number of free electrons/atom = 1.

To Find :

Fermi energy at Zero Kelvin, $E_{F_0} = ?$

Solution :

We have the equation for electron concentration,

$$n = \frac{\text{Number of free electrons per atoms} \times N_A \times D}{A},$$

$$= \frac{1 \times 6.025 \times 10^{26} \times 10500}{107.9},$$

$$= 5.863 \times 10^{28}.$$

The Fermi energy is given by the equation,

$$E_F = \left(\frac{h^2}{8m}\right)\left(\frac{3}{\pi}\right)^{2/3} n^{2/3},$$

$$= \frac{\left(6.63 \times 10^{-34}\right)^2}{\left(8 \times 9.11 \times 10^{-31}\right)}\left(\frac{3}{\pi}\right)^{2/3}\left(5.863 \times 10^{28}\right)^{2/3},$$

$$= 5.85 \times 10^{-38} \times 1.5091 \times 10^{19},$$

$$= 8.83 \times 10^{-19} \text{ J}.$$

$$\therefore \quad E_{F_0} = 8.83 \times 10^{-19} \times 6.24 \times 10^{18} \text{ eV} = 5.5 \text{ eV}.$$

Example 6.

Example 9 :

The Fermi level in silver is 5.5 eV. What are the energies for which the probabilities of occupancy at 300 K are 0.99, 0.01 and 0.5. (Given 1 J = 6.24×10^{18} eV).

Data :

For silver, $E_F = 5.5$ eV,

Temperature, $T = 300$ K.

To Find :

The energies for which $f(E) = 0.99$, 0.01 and 0.5, are to be found out.

Let them be E_1, E_2 and E_3 respectively.

$$\therefore \quad E_1 = ?, \; E_2 = ? \text{ and } E_3 = ?$$

Solution :

We have the equation,

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}.$$

Example 7.

Find the temperature at which there is 1% probability that a state with an energy 0.5 eV above fermi energy is occupied. V.T.U., Dec 13, Jan 09, June 08, July 04, (OS) Jan 07

Data :

Probability, $f(E) = 1\% = 0.01$.

Energy above E_F, i.e., $(E - E_F) = 0.5 \text{ eV} = 0.5 \times 1.6 \times 10^{-19}$ J.

To find :

Temperature at which the above data holds good, $T = ?$

Solution :

We have the equation for the probability of occupation that a given energy state is occupied as,

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}. \quad \text{......(1)}$$

Here, $$\frac{E-E_F}{kT} = \frac{0.5 \times 1.6 \times 10^{-19}}{1.38 \times 10^{-23} \times T} = \frac{5797}{T}.$$

Substituting the above in Eq (1), and using the data we have,

$$0.01 = \frac{1}{e^{\frac{5797}{T}} + 1}.$$

$\therefore$ $$T = 1261.1 \text{ K.}$$

Example 1.

Lead has a superconducting transition temperature of 7.26 K. If initial field at 0 K is 50 × 10^3 Am^{-1}, calculate the critical field at 6 K. Model QP1-2023

Data: Transition temperature, $T_c = 7.26$ K

& $T = 6$ K

Field at 0K, i.e., $H_0 = 50 \times 10^3$ Am-1

To find: Critical Feild $H_c = ?$

Solution: We have, $$H_c = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$H_c = 50 \times 10^3\left[1 - \left(\frac{6}{7.26}\right)^2\right] = 50 \times 10^3[1 - 0.623]$$

$$H_c = 18.834 \times 10^3 \text{ Am}^{-1}$$

Example 2.

The superconducting transistion temperature of Lead is 7.26 K. The initial field at 0 K is 64 × 10^3 Amp m^{-1}. Calculate the critical field at 5 K.

Solution:

Given data:

Critical temperature $T_C = 7.26$ K

Critical Field $H_0 = 64 \times 10^3$ Amp m^{-1}

Temperature = 5 K

The Critical Field $$H_C = H_0\left[1 - \left[\frac{T}{T_C}\right]^2\right]$$

$$= 64 \times 10^3\left[1 - \left[\frac{5}{7.20}\right]^2\right]$$

$$= 64 \times 10^3 \times 0.5257$$

$$H_C = 33.644 \times 10^3 \text{ Amp m}^{-1}$$

Example 3.

A superconducting material has a critical temperature of 3.7 K in zero magnetic field and a critical field of 0.02 T at 0 K. Find the critical field at 3 K.

Data: Transition temperature, $T_c = 3.7$ K

& $T = 3$ K

Field at 0K, i.e., $H_0 = 0.02$ Tesla

To find: Critical Feild $H_c = ?$

Solution: We have, $$H_c = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$H_c = 0.02\left[1 - \left(\frac{3}{3.7}\right)^2\right] = 0.02[1 - 0.657]$$

$$H_c = 0.013 \text{ Tesla}$$

Example 4.

The critical field for lead is 1.2 × 10^5 A/m at 8 K and 2.4 × 10^5 A/m at 0 K. Find the critical temperature of the material.

Data: Critical field, $H_c = 1.2 \times 10^5$ A/m

& $T = 8$ K

Field at 0K, i.e., $H_0 = 2.4 \times 10^5$ A/m

To find: Critical Temperature $T_c = ?$

Solution: We have, $$H_c = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$\rightarrow 1.2 \times 10^5 = 2.4 \times 10^5\left[1 - \left(\frac{8}{T_c}\right)^2\right] \quad \rightarrow \quad \frac{1.2\times10^5}{2.4\times10^5} = 1 - \frac{64}{T_c^2}$$

$$\rightarrow \frac{64}{T_c^2} = 1 - 0.5 \quad \rightarrow \quad T_c^2 = \frac{64}{0.5}$$

$$T_c = 11.3 \text{ K}$$

Example 5.

The critical field for niobium is 1 × 10^5 amp/m at 8 K and 2 × 10^5 amp/m at absolute zero. Find the transition temperature of the element.

Data: Critical Field, $H_c = 1 \times 10^5$ A/m

& $T = 8$ K

Field at 0K, i.e., $H_0 = 2 \times 10^5$ A/m

To find: Critical Temperature $T_c = ?$

Solution: We have, $$H_c = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$\rightarrow 1 \times 10^5 = 2 \times 10^5\left[1 - \left(\frac{8}{T_c}\right)^2\right] \quad \rightarrow \quad \frac{1\times10^5}{2\times10^5} = 1 - \frac{64}{T_c^2}$$

$$\rightarrow \frac{64}{T_c^2} = 1 - 0.5 \quad \rightarrow \quad T_c^2 = \frac{64}{0.5}$$

$$T_c = 11.3 \text{ K}$$

Example6.

The critical fields for lead are 1.2 × 10^5 A/m & 3.6 × 10^5 A/m at 12 K & 10 K, respectively. Find its critical temperature & critical field at 0K & 3.2 K.

Data: Critical field , $H_c(T_1) = 1.2 \times 10^5$ A/m , $T_1 = 12$ K

& $H_c(T_2) = 3.6 \times 10^5$ A/m , $T_2 = 10$ K

To find: Critical Temperature $T_c = ?$ & feild $H_{0K} = ?$ & $H_{3\cdot 2K} = ?$

Solution: We have, $$H_c = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right] \rightarrow H_c(T_1) = H_0\left[1 - \left(\frac{T_1}{T_c}\right)^2\right]$$

$$\rightarrow 1.2 \times 10^5 = H_0\left[1 - \left(\frac{12}{T_c}\right)^2\right] \quad (1) \qquad \text{Also } 3.6 \times 10^5 = H_0\left[1 - \left(\frac{10}{T_c}\right)^2\right] \quad (2)$$

On dividing Eqn. (1) by (2),

$$\frac{1}{3} = \frac{1-\left(\frac{12}{T_c}\right)^2}{1-\left(\frac{10}{T_c}\right)^2} \rightarrow 1 - \left(\frac{10}{T_c}\right)^2 = 3 - 3\left(\frac{12}{T_c}\right)^2 \rightarrow \frac{432-100}{T_c^2} = 2 \rightarrow T_c = 12.88\text{K}$$

Put in Eqn. (1), $$1.2 \times 10^5 = H_0\left[1 - \left(\frac{12}{12.88}\right)^2\right] \rightarrow H_0 = \frac{1.2 \times 10^5}{1 - \left(\frac{12}{12.88}\right)^2} \rightarrow H_0 = 9.06 \times 10^5 A/m$$

Now for **T=3.2K**,

$$H_c = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right] \rightarrow H_{3.2} = 9.06 \times 10^5\left[1 - \left(\frac{3.2}{12.88}\right)^2\right] \rightarrow H_{3.2} = 8.5 \times 10^5 A/m$$